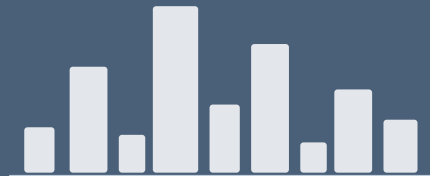


Interaction of a cumulus cloud ensemble with the large-scale environment, Part I

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J. Atmos. Sci. **31**, 674–701 (1974)

How do a spectrum of cumulus clouds and the large-scale flow control each other, and how can the clouds' collective effect be predicted?



Cumulus parameterization

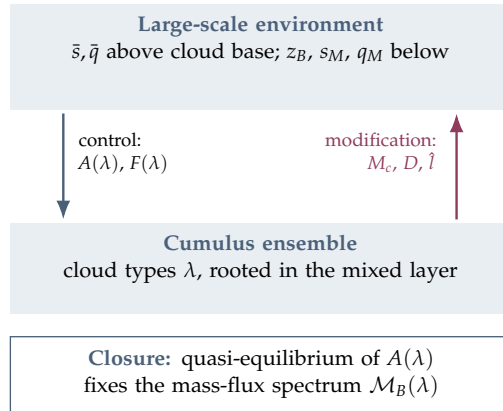
Many cumulus clouds live inside one large-scale disturbance, on much smaller scales. Predict only their collective effect, from large-scale variables.

Goal

A closed theory of the two-way interaction, with the subcloud mixed layer included.

Before this paper

CISK (Charney & Eliassen 1964) was too crude in general; Ooyama (1971) left the cloud spectrum open.



Closure: the clouds keep $A(\lambda)$ in balance with the large-scale forcing

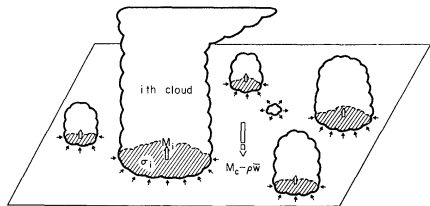


Fig. 1: a unit area between cloud base and the highest top. Tall clouds entrain; a cloud that lost buoyancy detrains.

$$M_i = \rho \sigma_i w_i, \quad M_c = \sum_i M_i, \quad \sigma_c = \sum_i \sigma_i \ll 1$$

$$s = c_p T + gz, \quad h = s + Lq$$

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{large scale}$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{large scale}$$

Detrainment: saturated cloud air with liquid water spreads out; its evaporation cools and moistens.

Cumulus-induced subsidence: warms and dries the environment between the clouds.

Condensation heating stays in the clouds. It keeps them buoyant and sustains M_c , so the environment warms through the subsidence, not directly.

The ensemble acts on the environment only through $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Cloud type λ : fractional entrainment rate, constant with height. Since $\mu = 2\alpha/R$, large λ means small clouds.

$$\partial_z \eta = \lambda \eta \Rightarrow \eta(z, \lambda) = e^{\lambda(z-z_B)}$$

$$h_c(z, \lambda) = \frac{1}{\eta} \left[h_M + \lambda \int_{z_B}^z \eta \bar{h} dz' \right]$$

Top $z_D(\lambda)$: buoyancy vanishes, $h_c = \hat{h}^* \approx \bar{h}^*$.

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

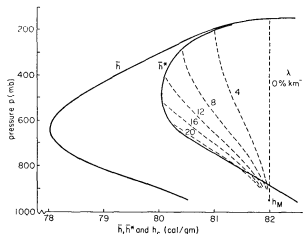


Fig. 6: $\bar{h}$, $\bar{h}^*$ and h_c for $\lambda = 0-20\% \text{ km}^{-1}$; tropical sounding.

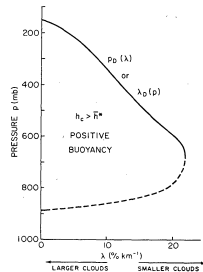


Fig. 8: top $p_D(\lambda)$.

In the paper

The clouds start from the subcloud mixed layer, $h_c(z_B, \lambda) = h_M$, whose depth z_B is predicted and is itself part of the control on convection.

Given the environment, each cloud type is fixed; only $\mathcal{M}_B(\lambda)$ is unknown

$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \frac{g}{c_p \bar{T}} \eta(z, \lambda) [s_{vc} - \bar{s}_v] dz$$

Kinetic energy generated per unit cloud-base mass flux, fixed by the environment alone. $\lambda = 0$: undiluted ascent, insensitive to the humidity above cloud base; larger λ also needs a moist environment.

$$\frac{dA}{dt} = \underbrace{\int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'}_{\text{clouds, } K < 0} + \underbrace{F(\lambda)}_{\text{large scale}}$$

Clouds consume their own instability; the large scale rebuilds it, mainly by cooling through ascent.

Quasi-equilibrium

$\tau_{ADJ} \sim 10^3$ – 10^4 s, the time the clouds need to remove A , against $\tau_{LS} \gtrsim 10^5$ s for the forcing. Then dA/dt is negligible:

$$\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) = 0$$

A Fredholm equation of the first kind for $\mathcal{M}_B \geq 0$: the spectrum of clouds is the one that balances the forcing.

In the paper

A sequence of quasi-neutral states, not a steady state: A stays small, not constant.

The clouds adjust fast enough that the large scale fixes their spectrum

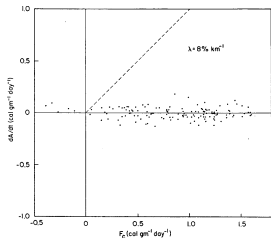


Fig. 13: observed dA/dt against the forcing F_C , Marshall Islands. Dashed: the change without clouds.

Observational check

$dA/dt \approx 0$ while F_C reaches $1.5 \text{ cal gm}^{-1} \text{ day}^{-1}$: the clouds consume the instability as fast as it is made.

Conclusions

The ensemble acts on the environment only through M_C , D and $\hat{I}$.

Split into entraining plumes, each cloud type is fixed by the environment; only $\mathcal{M}_B(\lambda)$ is left.

$A(\lambda)$ measures moist convective instability for each cloud type.

Quasi-equilibrium closes the theory: $\mathcal{M}_B(\lambda)$ solves a Fredholm equation of the first kind.

Limits

One parameter λ per cloud type; entraining updrafts only, no down-drafts.

Detrainment only at cloud top; crude rainfall; $\sigma_c \ll 1$; horizontally uniform environment; unsaturated mixed layer.

Only $\mathcal{M}_B(\lambda)$: no population, lifetime or cloud size. Triggering assumed, not modelled; shear neglected.

One tropical oceanic data set, without F_M ; the solution method is left to Part II.

A closed statistical theory of convection, tested on one tropical data set